

Tuesday Lichnerowicz Shift Derivation Prep — K-24 Cross-Check Framework

Lyra (Monday 2026-05-18 PM prep)

2026-05-18

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Purpose

Pre-stage the operator-translation derivation that bridges three distinct quantities for Tuesday's K-24 audit:

Quantity	Operator	Bundle	Form
Lyra $\text{Tr}(D^2_{\text{alg}} k)$	Algebraic Dirac D^2 at origin	Dolbeault spinor (rank 32)	$32 \cdot 10^k$ (closed, toy-verified)
Grace Coeff_n	Same Dirac D^2 (heat-kernel-coefficient)	Same spinor	$32 \cdot 10^n / n!$ (closed, toy-verified)
Elie a_n	Scalar Laplacian Δ on D_{IV}^5	Trivial line bundle (rank 1)	$(-1)^{\{n-1\} \cdot n! \cdot (n-1)! / (2 \{n-1\} \cdot n \cdot C^{\{n-1\}})}$ (Elie Toy 2994 closed)

These quantities are RELATED but NOT identical. Three operator/bundle distinctions separate them: 1. Operator-algebraic vs operator-full: D^2_{alg} (γ -matrix sums at origin, no gradients) vs D^2_{full} (with spin-connection gradient terms) 2. Spinor vs scalar bundle: spinor $\nabla^* \nabla$ on dim-32 Dolbeault bundle vs Δ on dim-1 scalar bundle 3. Local vs integrated: trace at origin (this morning's toy 3042) vs Seeley-DeWitt integral over D_{IV}^5 (Elie's a_n)

The Tuesday audit must respect these distinctions; direct numerical comparison without translation is invalid.

Explicit derivation steps (Tuesday morning work)

Step 1: Lichnerowicz formula on D_{IV}^5

Bergman scalar curvature on D_{IV}^5 : $R = -n_C \cdot g = -35$ (BST primary, T2352)

Lichnerowicz: $D^2_{\text{full}} = \nabla^* \nabla_{\text{spinor}} + R/4 = \nabla^* \nabla_{\text{spinor}} - n_C \cdot g/4 = \nabla^* \nabla_{\text{spinor}} - 35/4$

For constant scalar curvature (which holds on the Hermitian symmetric domain D_{IV}^5):

$\text{Tr}(e^{\{-t \cdot D^2_{\text{full}}\}}) = e^{\{+t \cdot n_C \cdot g/4\}} \cdot \text{Tr}(e^{\{-t \cdot \nabla^* \nabla_{\text{spinor}}\}})$ [constant shift factors out]

Step 2: Algebraic D^2_{alg} at origin vs full D^2_{full}

$D^2_{\text{alg}} = (\sum \gamma^{\{z_i\}} + \sum \gamma^{\{\bar{z}_j\}})^2$ (algebraic, at origin where Bergman metric is flat)

D^2_{full} has additional gradient/connection terms that contribute geometric curvature away from origin. At the origin itself, all gradient terms vanish identically (spin connection vanishes at base point on a Hermitian symmetric space — this is the local-rigidity statement of Helgason 1978).

At origin: $D^2_{\text{full}}|_{\{\text{origin}\}} = D^2_{\text{alg}}$ (no curvature corrections at the base point)

Away from origin: $D^2_{\text{full}} \neq D^2_{\text{alg}}$; the full operator has Bergman-metric-dependent terms.

Tuesday claim (to verify): the algebraic trace $\text{Tr}(D^2_{\text{alg}}^k)$ at origin equals the local Seeley-DeWitt coefficient at origin for D^2_{full} . This needs explicit verification via Hadamard-Minakshisundaram parametrix expansion.

Step 3: Spinor $\nabla^*\nabla$ vs scalar Δ

For the Dolbeault spinor bundle $S = \Lambda^* T^{\{0,1\}} D_{IV}^5$:

$\nabla^*\nabla_{\text{spinor}} = \Delta_{\text{scalar}} \otimes I_S + (\text{curvature-2-form terms from Chern character of } S)$

The “Chern character terms” are the BUNDLE corrections — they involve the curvature of the spinor bundle’s connection, which on D_{IV}^5 has explicit form via the Bergman-Kähler structure.

For a Hermitian symmetric space, the Atiyah-Singer index density of the Dolbeault complex on D_{IV}^5 relates the spinor heat-kernel trace to the scalar heat-kernel trace via:

$$\text{Tr}(e^{-t \cdot \nabla^* \nabla_{\text{spinor}}}) = \int_{D_{IV}^5} \text{ch}(S) \wedge \hat{A}(D_{IV}^5) \cdot K_t^{\{\text{scalar}\}}(x, x) \, d\text{vol}(x)$$

where $\text{ch}(S)$ is the Chern character of the spinor bundle, $\hat{A}(D_{IV}^5)$ is the A-roof genus, and $K_t^{\{\text{scalar}\}}$ is the scalar heat kernel.

For Type IV Hermitian symmetric domain D_{IV}^5 : $\text{ch}(S) = 2^{\{n_C\}} = 32$ (rank of spinor bundle), $\hat{A}(D_{IV}^5) = \hat{A}\text{-roof of } D_{IV}^5$.

Structural identification (predicted): $\text{Tr}(e^{-t \cdot \nabla^* \nabla_{\text{spinor}}}) = 32 \cdot \text{Tr}(e^{-t \cdot \Delta_{\text{scalar}}}) \cdot (\hat{A}\text{-correction})$

At leading order (flat origin): $\text{Tr}_{\text{origin}}(e^{-t \cdot \nabla^* \nabla_{\text{spinor}}}) = 32 \cdot \text{Tr}_{\text{origin}}(e^{-t \cdot \Delta_{\text{scalar}}})$

This is what I’ll verify Tuesday. The 32-fold multiplicity is the spinor-bundle dimension.

Step 4: Connection to Elie’s a_n

Elie’s a_n is the standard Seeley-DeWitt expansion of $\text{Tr}(e^{-t \cdot \Delta_{\text{scalar}}})$:

$$\text{Tr}(e^{-t \cdot \Delta_{\text{scalar}}}) \sim \sum_n a_n \cdot t^{-n} \quad (\text{small-}t \text{ asymptotic, integrated over } D_{IV}^5)$$

For the closed form Elie Toy 2994: $a_n = (-1)^{\{n-1\}n! \cdot (n-1)! / (2\{n-1\} \cdot n_C^{\{n-1\}})}$.

Per Steps 1-3 chain: $\text{Tr}(e^{-t \cdot D^2_{\text{alg}}})|_{\{\text{origin}\}} = 32 \cdot e^{-10t}$ (Lyra T2376 toy-verified) $\text{Tr}(e^{-t \cdot \nabla^* \nabla_{\text{spinor}}})|_{\{\text{origin}\}} = 32 \cdot e^{-\{-(10+n_C \cdot g/4)t\}} = 32 \cdot e^{-18.75t}$ (Lichnerowicz) $\text{Tr}(e^{-t \cdot \Delta_{\text{scalar}}})|_{\{\text{origin}\}} = \text{Tr}(e^{-t \cdot \nabla^* \nabla_{\text{spinor}}})/32$ (modulo $\hat{A}$ -correction) $= e^{-18.75t}$

BST primary form of the 18.75 shift (Grace T2377 compact decomposition): $18.75 = 75/4 = N_C \cdot n_C^2 / \text{rank}^2 = 3 \cdot 25/4$. Equivalently $75/4 = \text{rank} \cdot n_C + n_C \cdot g/4$ (my construction). Both decompositions hold; Grace’s form is more compact (three named primaries vs two-term sum). Filing both for completeness; canonical form per K51-style discipline favors Grace’s compact reading.

But this is the local pointwise heat kernel at origin, not the integrated Seeley-DeWitt coefficient that Elie’s a_n encodes. Elie’s a_n is the integrated version: $a_n = \int_{D_{IV}^5} a_n^{\{\text{local}\}}(x) \, d\text{vol}(x)$. The Bergman-volume integration introduces additional structure.

Per Keeper’s three-level framework (2026-05-18 PM clarification): my Step 1-3 chain bridges levels (1) point-trace algebraic $\leftrightarrow$ (2) point-trace heat-kernel-coefficient via Lichnerowicz. The bridge to level (3) integrated Seeley-DeWitt that Elie’s a_n encodes is multi-week — requires Bergman volume integration + $\hat{A}$ -correction. Keeper’s verdict: “Lichnerowicz works cleanly between (1)-Dirac and (1)-scalar. Doesn’t bridge (1) $\rightarrow$ (3) directly.” Tuesday audit operates at level (3); my pre-staged work delivers the level (1)/(2) operator translation. The level (3) test is whether Elie’s Three Theorems extension survives at $k=21..24$.

Critical Tuesday step: derive the volume-integration correction. The BST primary form should survive volume integration if D_{IV}^5 has a clean Bergman volume (which it does: $\text{Vol}(D_{IV}^5) = \pi^{\{n_C\}}/\text{something per Faraut-Koranyi}$).

Step 5: Falsifier test for Tuesday

If Elie's measured $a_{21..a_{44}}$ follow Lichnerowicz-shifted form of Grace's $32 \cdot 10^n/n!$ after appropriate operator/bundle/volume translations: - Cascade survives at 24 additional levels ($k=21..44$) - Structural-law promotion candidate (K52 follow-up via Keeper) - Paper #9 v11 content

If deviation at some k : - Locate cascade boundary (Casey's "deviations locate boundaries" principle) - Either Lichnerowicz factor is more complex than naive $R/4$ shift, or cascade has a boundary, or Dirac form needs corrections at higher k - Honest walk-back required; tier label adjusts

Honest scoping for Tuesday

What I will derive Tuesday morning: 1. Explicit Lichnerowicz shift between $\text{Tr}(D^2_{\text{alg}})$ and $\text{Tr}(\nabla^* \nabla_{\text{spinor}})$ at origin (D-tier, mechanical from $R = -n_C \cdot g$) 2. Spinor-bundle-dimension translation factor (D-tier, $32 = \text{rank}^{\{n_C\}}$) 3. Algebraic-vs-full operator distinction acknowledged with explicit claim about origin (verification of "all gradient terms vanish at origin" — needs check)

What is NOT yet closed by Tuesday morning: 1. Volume-integration translation from local origin trace to integrated Seeley-DeWitt (multi-week — requires Faraut-Koranyi Bergman volume decomposition + $\hat{A}$ -correction) 2. Numerical match of Elie's a_n to predicted BST primary form at sub-percent precision (depends on extracted values + multi-step translation) 3. Structural-law promotion (CONDITIONAL on cascade survival)

What Tuesday's audit ACTUALLY tests: - BST primary form preservation across operator translation - NOT direct numerical match (operators are different) - Structural-identification at each level: does the cascade pattern persist in Elie's data after appropriate translation, or does it break?

Tuesday audit deliverable plan

Step	Action	Owner	Outcome
1	Verify Elie $a_{2..a_{20}}$ reproduction from new checkpoint	Elie	Validation of extraction method
2	Derive Lichnerowicz shift (mechanical)	Lyra (Tuesday AM)	D-tier operator translation
3	Apply shift to Grace $32 \cdot 10^n/n!$ form	Grace (after Step 2)	Predicted Lichnerowicz-shifted scalar form
4	Compare Step 3 prediction vs Elie $a_{21..a_{44}}$	Keeper (after Steps 1-3)	Three-layer audit + K52-or-not ruling
5	Document outcome	Team trio	Paper #9 v11 candidate content OR honest walk-back

Pre-staged Tuesday claim

If Steps 1-4 land cleanly, the Tuesday audit will produce either: - Structural-law candidate for K52 promotion (Newton's G hierarchy cascade survives 24+ levels) - Honest walk-back identifying the cascade-boundary k (deviations locate boundary per Casey's principle)

Both outcomes are publishable. The reconciliation IS the audit, per Keeper's framing.

Audit-chain note

This prep document is Tuesday-morning derivation prep. It is NOT a structural-law claim yet. The Tuesday audit operates per the same discipline applied throughout Monday — honest tier labels, explicit operator distinctions, no premature promotion.

— Lyra, Tuesday prep filed 2026-05-18 ~13:15 EDT per Keeper's pre-audit reconciliation flag.